

INCIDENT INVESTIGATION REPORT

SOC Analyst Lab — Cybersecurity Detection & Response

Incident ID:	INC-2026-001	Report Date:	June 28, 2026
Analyst:	Cristian Benjamín Albuja Izurieta	Incident Start:	June 27, 2026
Classification:	CONFIDENTIAL — Lab Environment	Incident End:	June 28, 2026
Severity:	HIGH	Status:	RESOLVED

1. EXECUTIVE SUMMARY

Between June 27 and June 28, 2026, a controlled multi-stage attack simulation was conducted within a personal SOC laboratory environment targeting endpoint WIN-PQCPKD9AAN5 (192.168.64.20). The exercise was designed to replicate real-world adversary behavior and validate detection capabilities using Splunk Enterprise 10.4.0 as the SIEM platform.

Five MITRE ATT&CK; techniques were successfully simulated across the Execution, Discovery, Credential Access, Persistence, and Privilege Escalation tactics. All techniques generated corresponding Windows Security Event Log entries that were ingested, indexed, and detected within Splunk. Network traffic was simultaneously captured using Wireshark on a dedicated monitoring host (Kali Linux ARM64, 192.168.64.18).

A Python-based threat intelligence enrichment script was developed and executed to contextualize network indicators against AbuseIPDB and AlienVault OTX databases. All malicious artifacts were removed from the environment upon completion of the exercise. This report documents the complete investigation lifecycle from initial detection through remediation.

2. LAB ENVIRONMENT

Component	Details
Host Machine	MacBook Pro M3 — Apple Silicon ARM64 — 8GB RAM
Virtualization	UTM (QEMU) — Shared Network Mode
Target / SIEM Host	Windows 11 ARM64 — WIN-PQCPKD9AAN5 — 192.168.64.20
Monitoring Host	Kali Linux ARM64 — 192.168.64.18
SIEM Platform	Splunk Enterprise 10.4.0 — Index: win_logs — Port: 9997
Log Source	Windows Security Event Log (native) — Auditpol advanced audit
Network Analysis	Wireshark 4.6.6 — Interface: eth0
Threat Intel	Python 3.13 — AbuseIPDB API — AlienVault OTX API
Attack Simulation	Manual PowerShell execution — Native Windows tools

3. TIMELINE OF EVENTS

Date / Time	Event Description	EventCode	Severity
Jun 27 — Setup	Lab environment configured: Splunk installed, auditpol enabled, Wireshark deployed		INFO
Jun 28 18:36:30	PowerShell process with hidden window flag executed on target host	4688	HIGH
Jun 28 18:36:35	Network discovery commands executed: arp -a, ipconfig /all, net view	4688	MEDIUM
Jun 28 18:47:48	Multiple failed logon attempts detected — brute force pattern identified	4625	HIGH
Jun 28 19:27:09	Unauthorized scheduled task created: WindowsUpdateHelper (persistence)	4698	HIGH
Jun 28 19:27:09	New local user account created: attacker	4720	CRITICAL
Jun 28 19:27:09	Account 'attacker' added to local Administrators group	4732	CRITICAL
Jun 28 23:15:34	Threat intelligence enrichment performed — IPs queried against AbuseIPDB/OTX		INFO
Jun 28 23:50:00	Malicious artifacts removed — environment restored to clean state	—	RESOLVED

4. ATTACK NARRATIVE

The simulated attack followed a structured multi-stage intrusion chain representative of real-world adversary behavior observed in enterprise environments.

Stage 1 — Execution (T1059.001)

The threat actor initiated the attack by executing PowerShell with evasion flags (-nop -w hidden), a technique commonly associated with malware dropper stages and command-and-control beacon establishment. This generated EventID 4688 (new process created) in the Windows Security log.

Stage 2 — Discovery (T1018)

Following initial execution, network reconnaissance was conducted using native Windows tools (arp -a, ipconfig /all, net view /all) to enumerate hosts and network configuration. This living-off-the-land technique avoids introducing external tooling and is difficult to distinguish from legitimate administrator activity without behavioral baseline analysis.

Stage 3 — Credential Access (T1110)

A brute force attack was simulated against local account credentials, generating five consecutive EventID 4625 (failed logon) entries. This pattern triggers detection rules in environments monitoring for account lockout thresholds.

Stage 4 — Persistence (T1053.005)

To establish persistence, a scheduled task named 'WindowsUpdateHelper' was created to execute a hidden PowerShell payload at every user logon under the SYSTEM account. This technique survives reboots and is commonly used by ransomware and APT groups for maintaining access. EventID 4698 (scheduled task created) was generated and detected in Splunk.

Stage 5 — Privilege Escalation (T1078)

A new local administrator account named 'attacker' was created using net user commands and immediately added to the local Administrators group. This provides the threat actor with persistent elevated access independent of the compromised account, detected via EventID 4720 (user account created).

5. MITRE ATT&CK; MAPPING

Technique ID	Technique Name	Tactic	Detection Event	Result
T1059.001	PowerShell Execution	Execution	EventID 4688 — hidden flag	Detected
T1018	Remote System Discovery	Discovery	EventID 4688 — recon commands	Detected
T1110	Brute Force	Credential Access	EventID 4625 — failed logons	Detected
T1053.005	Scheduled Task/Job	Persistence	EventID 4698 — task creation	Detected
T1078	Valid Accounts	Privilege Escalation	EventID 4720 — account creation	Detected

6. INDICATORS OF COMPROMISE (IoCs)

6.1 Network Indicators — Threat Intelligence Enrichment

Network indicators identified during the investigation were submitted to threat intelligence platforms via automated Python script. Results are documented below.

IP Address	Type	AbuseIPDB Score	OTX Pulses	ISP	Verdict
192.168.64.1	Gateway — Private	0/100	0	N/A (RFC1918)	CLEAN
8.8.8.8	Public DNS	0/100 (121 historical)	0	Google LLC — US	CLEAN

6.2 Host-Based Indicators

Artifact Type	Value / Name	Associated Technique
Malicious User Account	attacker (local administrator)	T1078 — Valid Accounts
Unauthorized Scheduled Task	WindowsUpdateHelper	T1053.005 — Scheduled Task
Suspicious Process	powershell.exe -nop -w hidden -c ...	T1059.001 — PowerShell
Recon Command Set	arp -a / ipconfig /all / net view /all	T1018 — System Discovery
Brute Force Target	fakeuser — 5 failed logon attempts	T1110 — Brute Force

7. SPLUNK DETECTION QUERIES (SPL)

All queries executed against index=win_logs. Time range: June 27–28, 2026.

7.1 T1059.001 — PowerShell Execution with Evasion Flags [Severity: HIGH]

```
index=win_logs EventCode=4688 | search CommandLine="*powershell*" OR  
CommandLine="*-nop*" OR CommandLine="*-w hidden*" | table _time, NewProcessName,  
CommandLine | sort -_time
```

7.2 T1110 — Brute Force / Multiple Failed Logons [Severity: HIGH]

```
index=win_logs EventCode=4625 | stats count by TargetUserName, IpAddress | where count  
> 3 | sort -count
```

7.3 T1053.005 — Unauthorized Scheduled Task Creation [Severity: HIGH]

```
index=win_logs EventCode=4698 | table _time, TaskName, TaskContent | sort -_time
```

7.4 T1078 — New User Account Creation [Severity: CRITICAL]

```
index=win_logs EventCode=4720 | table _time, TargetUserName, SubjectUserName | sort -_time
```

7.5 T1018 — Network Discovery Commands [Severity: MEDIUM]

```
index=win_logs EventCode=4688 | search CommandLine="*arp*" OR CommandLine="*ipconfig*" OR CommandLine="*net view*" | table _time, CommandLine | sort -_time
```

8. IMPACT ASSESSMENT

Category	Finding	Severity
Confidentiality	Simulated credential exposure through unauthorized admin account creation	HIGH
Integrity	Unauthorized scheduled task established for persistence across reboots	HIGH
Availability	No service disruption detected during the simulation exercise	LOW
Lateral Movement	Admin account 'attacker' could facilitate lateral movement to other hosts	CRITICAL
Data Exfiltration	No exfiltration detected; all network traffic analyzed via Wireshark capture	LOW

9. RESPONSE ACTIONS

9.1 Containment

- Identified affected endpoint WIN-PQCPKD9AAN5 and suspended outbound activity
- Terminated suspicious PowerShell processes detected during simulation
- Windows Defender real-time monitoring was re-enabled post-exercise

9.2 Eradication

- Deleted malicious user account: net user attacker /delete
- Removed unauthorized scheduled task: schtasks /delete /tn WindowsUpdateHelper /f
- Registry and service entries verified clean post-removal

9.3 Evidence Preservation

- Splunk search results exported for all five detected MITRE ATT&CK; techniques
- Wireshark network capture saved: attck_simulations.pcapng (Kali Desktop)
- Threat intelligence enrichment output saved: threat_intel_results.json
- Windows Security Event Log entries preserved for forensic documentation
- Python threat intel script archived: threat_intel_lookup.py

10. RECOMMENDATIONS

1. Enable PowerShell Script Block Logging (EventID 4104)

Configure Group Policy to capture full PowerShell command content including encoded payloads, providing deeper visibility than process creation events alone.

2. Deploy Sysmon for Enhanced Host Telemetry

Install Sysmon with SwiftOnSecurity configuration to capture network connections (EventID 3), file creation (EventID 11), and process injection activity unavailable in native Windows logging.

3. Implement Real-Time Alerts for Privilege Events

Configure Splunk alerts for EventID 4720 (new account creation) and EventID 4732 (account added to privileged group) to detect unauthorized privilege escalation within minutes.

4. Enforce Scheduled Task Auditing Policy

Enable Object Access auditing and create Splunk alerts on EventID 4698 for any scheduled task creation outside approved maintenance windows.

5. Apply Network Segmentation Controls

Implement network segmentation to limit lateral movement. Critical assets should require explicit multi-factor authentication for inter-segment access.